Sarvesh Baskar

Dual Degree Student (B.E. Computer Science & M.Sc. Physics), BITS Pilani
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RESEARCH INTERESTS

Large Language Models (LLMs), Reinforcement Learning, Reasoning & Planning, Robotics, Machine Perception, Computer Vision, Vision-Language Action Models.

EDUCATION

Birla Institute of Technology and Science (BITS) Pilani — Sep 2020 to July 2025
Dual Degree:
Bachelor of Engineering in Computer Science — Cumulative GPA: 8.46/10.00
Master of Science in Physics
Thesis Title: An Approach to Resolving the Persona Knowledge Gap in LLMs during Multi-Turn Conversations (Link)

PUBLICATIONS

Journal/Conference Papers
Published

- Sarvesh Baskar, Manas Gaur, Srinivasan Parthasarathy, and Tanmay Tulsidas Verlekar. 2025. (CPER) From Guessing to Asking: An Approach to Resolving Persona Knowledge Gap in LLMs during Multi-Turn Conversations. In Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 4: Student Research Workshop), pages 435–447, Albuquerque, USA. Association for Computational Linguistics. (*Link*)

ACADEMIC EXPERIENCE

University of Maryland, College Park — Remote
Research Assistant (Furong's Lab) — July 2025 to Present

- Collaborate with PhD students and a postdoctoral researcher to scope tasks and plan experiments.
- Run controlled experiments and ablations; track configurations, metrics, and results for reproducibility.
- Read Literature and synthesize relevant papers to inform methods, datasets, and evaluation.
- Deliver concise weekly updates with findings, figures, and next steps.

University of Maryland, Baltimore County (UMBC) — MD, USA
Research Assistant (KAI2 Lab) — Aug 2024 to Jul 2025
Advisors: Dr. Manas Gaur, Dr. Rajasekhar Anguluri

- First-author NAACL 2025 paper on conversation preference elicitation for LLMs; led experiments, analysis, and writing.
- Designed and ran ablations for planning agents and RL for LLMs; converted literature insights into concrete experiment plans and evaluation protocols.
- Extended **ATLAS** (Automated Thought of Search) to harder ACPBench classical planning domains in collaboration with IBM; defined interfaces, added benchmarks, and standardized analysis.
- Co-wrote sections of the Samsung START 2025 proposal, including scope, methodology, milestones, and expected outcomes.
- Built reproducible experiment harnesses: data pipelines, configuration schemas, SLURM scripts, and result logging for faster iteration.

TCS Research - Datalab, APPCAIR, BITS Pilani KK Birla Goa Campus — Goa, India
Student Researcher — Jan 2024 to May 2024

- Conducted comprehensive literature reviews to scope tasks, identify baselines, datasets, and evaluation protocols.
- Translated high level research goals into concrete experimental plans and weekly milestones with professors and industry collaborators.
- Implemented the discussed methodology pipelines in Python with version control, unit tests, and clear documentation for reproducibility.
- Prepared weekly written reports and presentations that summarized results, insights, and next steps for faculty and industry stakeholders.
- Coordinated code handoff by organizing repositories, and packaging notebooks for maintainability and reuse.
- Contributed to manuscript sections on related work, methods, experimental setup, and result interpretation.

Birla Institute of Technology and Science (BITS) Pilani, KK Birla Goa Campus Goa, India
Teaching Assistant Aug 2023 to May 2024

- Assisted instruction for Mathematical Methods of Physics (Fall 2023–24) and Computational Physics (Spring 2023–24)
- Graded assignments, problem sets, and lab exercises.
- Conducted demonstrations for numerical methods (e.g., integration, differential equations) and scientific computing best practices.
- Supported exam evaluation; provided feedback to students and instructors to improve learning outcomes.

PROJECTS

ATLAS: Automated Thought Of Search for Language Described Search Problems MD, USA/ Remote
Collaborative Project with IBM and UMBC May 2025 to Present
Collaborators: Harsha Kokel, Michael Katz, Kavitha Srinivas, Shirin Sohrabi, Manas Gaur

- **Problem.** Natural-language described search problems are underspecified and hard to ground into executable search procedures. Pure LLM prompting is brittle and lacks verification, which leads to incorrect plans and poor generalization across domains.
- **What I built.** Designed *ATLAS*, a modular thought-of-search pipeline that parses NL tasks into structured specs, synthesizes components with a unit-test feedback loop, and runs a validator-guided breadth-first search over candidate programs and plans. Implemented an ablation suite and evaluation harness to isolate the contribution of tests, feedback, and search.
- **Impact and Results.** ATLAS solved **149/160** tasks (**93.1%**) across 8 classical planning domains (blocksworld, ferry, floortile, visitall, grippers, grid, rovers, logistics), versus strongest ablation **88/160** (**55.0%**) and base model with **102/160** (**64.0%**).

VisAlign - Mitigating Hallucinations via Refining Textual Embeddings MD, USA / Remote
Ongoing Collaborative Project with (Ph.D. candidate) Aakriti Agrawal from UMD Oct 2025 to Present

- **Problem:** Large vision language models tend to over rely on language tokens because visual embeddings are simply appended to the text stream, which leads to weak visual grounding and hallucination.
- **What I did:** Implemented the VisAlign refinement in the OpenQwen2VL stack and ran the full pretraining and evaluations. Integrated average pooled visual features into textual embeddings at inference, executed experiments, and prepared the result tables and plots for the paper.
- **Impact and Results.** Integrated VisAlign into OpenQwen2VL and ran full pretraining plus eval scripts. Preliminary benchmarks suggest lower hallucination rates with improved grounding; full OpenQwen2VL numbers pending.

MORSE-PIC MD, USA / Remote
Ongoing Collaborative Project with (PostDoc) Dr. Zikui Cai from UMD Sep 2025 to Present

- **Problem:** Self improvement of smaller vision models by generating difficulty controlled variants of an unsolved visual reasoning task and fine tuning on them. Stress testing of larger models by increasing task difficulty to find failure thresholds and map robustness.
- **What I did:** An end to end coding agent pipeline that, given a problem the model fails, programmatically synthesizes same kind images with tunable difficulty and routes curated sets to fine tuning and to stress testing.
- **Impact and Results.** The agent reliably produces 2D synthetic image families that are suitable for training and evaluation; 3D scene recreation is not yet reliable and is a known limitation. Next step: fine tune smaller models on the synthetic curriculum to demonstrate measurable self-improvement

Multi-Step Reasoning Using Process Reward Models MD, USA / Remote
Collaborative Project with (Ph.D. candidate) Souradip Chakraborty from UMD July 2025 to Aug 2025

- **Problem:** Outcome-only supervision in math reasoning rewards the final answer but ignores the quality of intermediate steps. This leads to brittle chains of thought, unnecessarily long solutions, and failures that are hard to diagnose or correct. Our hypothesis is that using PRM would produce more correct steps.
- **What I built:** An end-to-end PRM pipeline that generates multi-step trajectories, assigns weak step-level labels using verifier/self-consistency signals, and uses that score at inference for PRM-guided answers.
- **Impact and Results.** Experimenting on GSM8K PRM guidance *underperformed* both baselines across held out sets. Our initial results demonstrates that Majority voting was consistently strongest with a 92% accuracy compared to PRM with a 68% accuracy.

Large Language Models for Code Deobfuscation MD, USA
Collaborative project with (Ph.D. candidate) Seyedreza Mohseni from UMBC Feb 2025 to May 2025

- **Problem:** Automating deobfuscation of low level LLVM-IR is difficult with existing tools that struggle under Bogus Control Flow and Control Flow Flattening, and there is no standardized AI evaluation for this setting.

- **What I did:** Led the AI modeling and evaluation track. Designed Chain of Thought procedural prompts with global variable tracing. Built an LLVM-IR analysis module to extract basic blocks and control flow graphs and to compute Cyclomatic Complexity and instruction counts. Implemented an adjustment step to normalize identifiers for fair Levenshtein similarity. Created an evaluation harness using Code Reduction, Levenshtein Similarity, and Cyclomatic Complexity and benchmarked four reasoner models on O-LLVM obfuscations.
- **Impact and Results.** Established a 36 sample LLVM-IR benchmark across BCF, CFF, and BCF-CFF. Best model achieved **76.9%** code reduction and **85%** similarity on BCF-CFF with Cyclomatic Complexity reduced from **91** to **13.5** (**85%**).

From Guessing to Asking (CPER) (Link) — MD, USA
Thesis — Aug 2024 to Dec 2024

- **Problem:** Large language models in multi-turn chats miss user-specific context, creating a *persona knowledge gap* that leads to generic, incoherent, or misaligned replies.
- **What I did:** Designed *CPER*, a three-module framework that quantifies uncertainty, asks targeted clarifying questions, and generates persona-aligned responses. Implemented and evaluated on CCPE-M and ESConv.
- **Impact and Results.** Human A/B judges preferred CPER over strong baselines by **+42%** on CCPE-M and **+27%** on ESConv, with strongest gains in long (12+ turn) conversations; GPT-4o preference on CCPE-M reached **60.78%**. Published at *NAACL 2025*.

Automated Planning and Code Synthesis for Continual ML Ideation (Link) — Goa, India
Project under Datalab, APPCAIR - TCS Research — Jan 2024 to May 2024
Collaborators: Dr. Gautam Shroff, Dr. Lovekesh Vig, Prof. Ashwin Srinivasan, Prof. Tanmay Tulsidas Verlekar

- **Problem.** Turning new research ideas into runnable changes in an existing ML pipeline is slow and inconsistent. Teams need automatic extraction of ideas from papers and automatic plan to code execution with minimal manual effort.
- **What was built.** An LLM driven system that mines techniques and goals from a set of papers with a RAG workflow, then produces a modification plan, and generates, debugs, and finalizes code.
- **Impact and Results.** Internal runs showed about a **30% increase in executable code yield** from optimized prompting and iterative repair, and successful end to end execution of planned modifications with minimal manual intervention.

INDUSTRY EXPERIENCES

Avyott — Goa, India / Remote
AI Consultant — Aug 2025 to Nov 2025

- Designed and shipped a multimodal RAG chatbot embedded inside client products, able to process text, images, and tables for grounded and cited answers.
- Wrote and deployed Model Context Protocol (MCP) servers to connect with multiple ticketing systems, enabling chat driven ticket creation, updates, and status sync.
- Implemented a clarification step that asks targeted questions before answering, based on my thesis research, which improved personalization and intent capture.
- Deployed the product for enterprise use at IPCA and Taj, integrating with internal knowledge bases and existing workflows.

Techisy — Goa, India / Remote
Summer Intern — May 2024 to July 2024

- Built an AI-powered duplicate bill identification system (95% accuracy) combining OCR, OpenAI embeddings, and CLIP, reducing manual review time.
- Created a RAG-based resume matching solution using LangChain, Ollama (Llama 3), and FAISS to align resumes with job descriptions.
- Deployed prototypes and iterated with stakeholders to improve screening efficiency.

SKILLS

- **Programming:** Python, C/C++, LaTeX, Bash, Slurm
- **Software/Tools:** PyTorch, LangChain, DSPy, Git, TensorFlow, OpenCV, vLLM, Redis, ChromaDB
- **Soft Skills:** Research writing, Teaching/mentoring, Communication, Collaboration, Problem solving

ENGLISH TESTS

IELTS (Academic): Overall: 7.5 | Listening: 7.5 | Reading: 7.5 | Speaking: 8.0 | Writing: 7.0